

Can Demircan

Phd student interested in LLM interpretability, AI alignment, and representation learning.

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🔗 Scholar

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Education

- 2023 – ... **Ph.D., Technical University of Munich** in Machine Learning & Cognitive Science.
Supervisors: Dr. Eric Schulz (Helmholtz Munich) & Prof. Dr. Zeynep Akata (TUM).
Topics: AI alignment, LLM interpretability, and representation learning.
- 2021 – 2023 **M.Sc., University of Tuebingen** in Neural & Behavioural Sciences (Final Grade: 1.3).
Thesis: *Using tools of neuroscience to understand large language models.*
- 2017 – 2020 **B.A., Wadham College, University of Oxford** in Experimental Psychology.
Graduated with 1st Class Honours

Experience

- 2022 – ... **Instructor with the Software Carpentry.**
Taught and helped with workshops on Python, Git, and Bash at the University of Tuebingen and the University of Twente.
- 2021 – 2023 **Research Assistant** at the Max Planck Institute for Biological Cybernetics.
Investigated the representational basis of human learning in naturalistic tasks using online experiments and computational modelling.

Skills

- Technical **Python, R, HTML/CSS/JS/jsPsych, Git, Bash, Docker/Singularity, CI/CD, $\LaTeX$.**
- Research **Multi-device model training in PyTorch, online behavioural experiments, cognitive and statistical modeling, verbal, written, & visual communication of research.**
- Languages **Turkish (native), English (fluent), German (beginner-intermediate).**

Awards & Grants

- 2025 **Lambda Research Grant** \$1000 worth of cloud compute for interpretability research on language models trained to capture human cognition.
- 2021 – 2023 **International Max Planck Research School Stipend** Two years of full funding to pursue an M.Sc. degree in Tuebingen.

Peer-Reviewed Publications & Preprints

- 2026 **Demircan, C.,** Binz, M., Modirshanechi, A., & Schulz, E. Meta-learning as a principle for human-like visual representations. *Under Review.* [\[paper\]](#) [\[code & data\]](#)
- Schulze Buschoff, L. M.,** Voudouris, K., **Demircan, C.,** & Schulz, E. Can Vision Language Models Learn Intuitive Physics from Interaction? *International Conference on Machine Learning (ICML).* [\[paper\]](#)
- 2025 **Saanum, T.*, Demircan, C.*,** Gershman, S. J., & Schulz, E. A circuit for predicting hierarchical structure in-context in Large Language Models. *Under Review.* [\[paper\]](#) [\[code\]](#)
- Binz, M., ..., Demircan, C., ...,** Schulz, E. A foundation model to predict and capture human cognition. *Nature.* [\[paper\]](#)
- Demircan, C.*,** Saanum, T.*, Jagadish, A. K., Binz, M., & Schulz, E. Sparse Autoencoders Reveal Temporal Difference Learning in Large Language Models. *International Conference on Learning Representations (ICLR).* [\[paper\]](#)

Peer-Reviewed Publications & Preprints (continued)

- 2024  **Demircan, C.**, Saanum, T., Pettini, L., Binz, M., Baczkowski, B. M., Doeller, C., Garvert, M. M., & Schulz, E. Evaluating alignment between humans and neural network representations in image-based learning tasks. *Advances in Neural Information Processing Systems (NeurIPS)*. [\[paper\]](#) [\[code\]](#) [\[data\]](#)
-  Özdemir, Ş., Şentürk, Y. D., Ünver, N., **Demircan, C.**, Olivers, C. N. L., Egner, T., & Günseli, E. Effects of context changes on memory reactivation. *Journal of Neuroscience*. [\[paper\]](#)
-  Şentürk, Y. D., Ünver, N., **Demircan, C.**, Egner, T., & Günseli, E. The reactivation of task rules triggers the reactivation of task-relevant items. *Cortex*. [\[paper\]](#)

Non-Archival Publications

- 2025  Naranjo, I., **Demircan, C.**, Schulz, E. How Does an LLM Process Conflicting Information In-Context? 8th Annual Conference on Cognitive Computational Neuroscience (CCN). [\[paper\]](#)
- 2022  **Demircan, C.**, Pettini, L., Saanum, T., Binz, M., Baczkowski, B. M., Doeller, C., Garvert, M. M., Schulz, E. Decision-Making with Naturalistic Options. In *Proceedings of the Annual Meeting of the Cognitive Science Society (Vol. 44, No. 44)* [\[paper\]](#) [\[code & data\]](#)

Supervision

- 2024 – 2025  **Ivan Naranjo** B.Sc. in Computer Science, Technical University of Munich.
Thesis: *How Does an LLM Process Conflicting Information In-Context?*

Reviewing

- 2026  International Conference on Machine Learning (ICML)
- 2025  International Conference on Learning Representations (ICLR)
-  Conference on Neural Information Processing Systems (NeurIPS)
-  NeurIPS Workshop on CogInterp: Interpreting Cognition in Deep Learning Models
-  Conference on Cognitive Computational Neuroscience (CCN)